Contrastive Learning Based on Category Matching for Domain Adaptation in Hyperspectral Image Classification

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***Abstract*— Cross-scene hyperspectral image classification (HSIC) is a challenging topic in remote sensing, especially when there are no labels in the target domain. Domain adaptation (DA) techniques for cross-scene HSIC aim to label a target domain by associating it with a labeled source domain. Most existing DA methods learn domain-invariant features by reducing feature distance across domains. Recently, contrastive learning has shown excellent performance in computer vision (CV) tasks, but there is little or no research on the performance of cross-scene HSIC. Considering that its idea is similar to reducing feature distance, this article attempts to explore whether contrastive learning can achieve cross-scene HSIC. In this work, an instance-to-instance contrastive learning framework based on category matching (CLCM) is designed. The main idea is to take the category information as the premise in the feature space, regard the source sample as an anchor, and find its positive and negative matching samples across domains. The instance-level discriminative feature embeddings are learned through positive matching pairs attracting each other and negative matching pairs repelling each other. Among them, the target label is a pseudo-label. To further improve the quality of contrastive learning, it is considered to focus on extracting the spectral–spatial features of hyperspectral images (HSIs) to more accurately represent semantic information. Simultaneously, high-confidence target samples are screened to update the network. Three DA tasks confirm the effectiveness and feature discriminativeness of CLCM, while also providing new ideas for cross-scene image classification.**

***Index Terms*— Category matching, contrastive learning, domain adaptation (DA), hyperspectral image classification (HSIC).**

I. Introduction

With the successful launch of many remote-sensing satellites, various remote-sensing images become available [1]. Hyperspectral image (HSI) is an important data type in remote sensing, containing hundreds of narrow spectral bands. Due to the rich spectral information, it has been widely used in many fields, such as environmental monitoring [2], [3], mineral exploration [4], and land cover classification [5]. HSI classification (HSIC) aims to classify each pixel in HSIs, which is a key technique in practical applications. However, HSIC is very difficult due to the large amount of data, high dimensionality, and difficulty in obtaining labels [2]. Although different small-sample HSIC methods are proposed to solve the label scarcity problem, the models trained on limited labeled samples are usually unstable and lack of generalization ability. When HSI has no labels, these supervised classification methods cannot be used. One feasible way to perform classification of unlabeled HSI (called target domain) is to use existing related and similar labeled HSI (called source domain) by means of model transfer or domain adaptation (DA) techniques [6], [7], [8]. Due to variations of acquisition conditions, such as different sensors, different times, different regions, and so on, there will be a large spectral shift between two images from different domains [9], [10], [11], [12]. Therefore, the traditional idea of training a model on an image (source domain) and applying it directly to the image that needs to be classified (target domain) cannot achieve satisfactory results [8].

DA technique can solve the spectral shift or domain shift between domains and assign labels to target domains via related labeled source domains [7], [8]. In recent years, many DA methods have been proposed. Roughly, traditional DA methods can be divided into three types: instance-based, classifier-based, and feature-based [8]. The idea of instance-based methods is to reweight samples or extract important instances to reduce domain shift [8], [13]. Classifier-based methods aim to train a robust classifier that can then induce good generalization to the target domain. Feature-based methods are the most common ones, which intend to extract domain-invariant features to reduce the distribution difference between two domains [12], [14], [15]. In the beginning, traditional DA methods, such as subspace alignment (SA) [16], correlation alignment (CORAL) [17], geodesic flow kernel (GFK) [18], transfer component analysis (TCA) [19], joint distribution adaptation (JDA) [20], discriminative transfer joint matching (DTJM) [21], all learn efficient transferable feature representations by learning linear transformations for SA or distribution alignment.

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However, these methods can only extract shallow hand-crafted features.

With the introduction of deep learning, due to its ability to automatically extract features, all deep DA methods have emerged, reflecting better advantages. Long et al. [22] proposed a deep adaptation network (DAN) to embed representations into a common space to reduce domain shift by minimizing the multikernel maximum mean discrepancy (MMD). In addition to methods that focus on global domain shift, Zhu et al. [23] considered the alignment between two subdomains within different domains of the same category. To this end, a deep subdomain adaptation network (DSAN) was proposed, which applies local MMD to align the subdomain distribution to train the network. Since a single structural feature extraction may contain partial information, along this line of thought, Zhu et al. [24] proposed a multirepresentation adaptation network (MRAN). The network extracts information from different perspectives for multirepresentation alignment and uses the extended conditional MMD to measure the domain discrepancy [24]. Note that quite a few approaches achieve domain alignment by optimizing statistical distribution variance, that is, MMD and its variants. For cross-scene HSIC, Fang et al. [25] proposed confident learning-based DA (CLDA) to achieve cross-domain HSI alignment. Specifically, it uses confident learning to find high-confidence target samples, increasing the discriminative ability of neural networks. Simultaneously training with confident learning and DA can further improve classification performance. Zhang et al. [14] proposed a topological structure and semantic information transfer network (TSTnet), using graph convolutional networks (GCNs) to improve the spatial perception ability of HSI, and using graph optimal transmission (GOT) and MMD for graph alignment and distribution alignment, respectively. In addition, consistency constraints are used to coordinate the two alignments.

Recently, self-supervised learning, such as contrastive learning, is proposed to learn feature representations at the instance level, which has been proven effective in the field of computer vision (CV) [26], [27]. The idea of contrastive learning is to pull the picture onto the hypersphere closer to its own augmented copy and away from other pictures. Since in standard contrastive learning, the enhanced image associated with a certain image (anchor) is considered, it is natural to generate positive matching samples. However, in the DA problem, it is not clear how to generate anchor-related matching samples. After all, there are two domains to consider, and there are differences between them. Currently, some works extend traditional contrastive learning, that is, learning in a single domain, to bridge domain differences by performing contrastive learning across domains. For example, Wang et al. [28] proposed a simple cross-domain contrastive learning (CDCL) framework. Given an anchor image from one domain, CDCL aims to minimize its distance to cross-domain samples from the same class. It uses source and target domains as anchors, respectively, to construct two-way contrastive learning. Considering that target labels are not available, K-means clustering is used to initialize the pseudo-labels in the target domain [28]. Park et al. [29] introduced joint contrastive learning (JCL) to discover discriminative features to distinguish each class. Under this framework, samples from two domains are jointly optimized for minimal joint error [29]. Specifically, a contrastive loss is used to enforce the close correspondence between features and labels, and Jensen–Shannon (JS) difference is used to enlarge the conditional distribution of different classes in mixed datasets. In addition, Singh [30] proposed a contrastive learning framework for semisupervised DA (CLDA). On the one hand, the similarity between homogeneous centroids across domains is maximized by implementing interdomain contrastive alignment. On the other hand, intradomain differences are reduced by implementing instance-level contrastive learning between unlabeled target images and their strongly enhanced images [30].

The aforementioned methods of combining contrastive learning and DA have a good effect in the field of CV, but there is no or little research on the performance of cross-scene HSIC. After all, cross-scene HSIC is fundamentally different from natural image classification. Each natural image is a sample, and it is more natural to construct positive matching samples through data enhancement and then perform contrastive learning. Operations, such as rotation, cropping, and so on, are all meaningful. However, HSIs do not have certain properties of natural images, where one pixel represents one sample. So, direct application or extension cannot work well. In unsupervised DA tasks, however, the idea of contrastive learning is very similar in spirit to reducing feature distribution distances. Therefore, considering that contrastive learning is simple and feasible for domain shift, this article intends to fit contrastive learning to cross-domain HSIC according to the characteristics of HSIs. To this end, a contrastive learning framework based on category matching (CLCM) is proposed to perform instance-to-instance contrastive learning. Specifically, category matching pairs are constructed across domains, so that instances of the same (different) class become positive (negative) matching pairs, regardless of which domain the instances come from. Then, the robust and class-discriminative representation of the two domains is enhanced by minimizing (maximizing) the distance of positive (negative) matching pairs.

To further improve the quality of contrastive learning, it is expected that feature representation of instances can be more accurate and effective, and the category information has higher reliability. For HSIs, to effectively integrate spatial and spectral information, 2-D and 3-D convolutional neural networks (CNNs) are applied to extract deep representations of HSIs. For example, Zhong et al. [31] proposed a spectral–spatial residual network (SSRN) to learn deep discriminative features from HSI using multiple spectral and spatial residual blocks. Roy et al. [32] designed a hybrid spectral CNN (HybridSN) to extract the hybrid spectral–spatial features of HSIs by using the sequence of first 3-D convolution and then 2-D convolution. Therefore, the proposed CLCM's feature extraction network also mainly uses 3-D convolution as the basic element to extract the feature representation of the pixels in HSI, compared with the common ResNet-50 and so on. Aiming at the credibility of the category information, considering the pseudo-labels obtained by the task-specific classification layer

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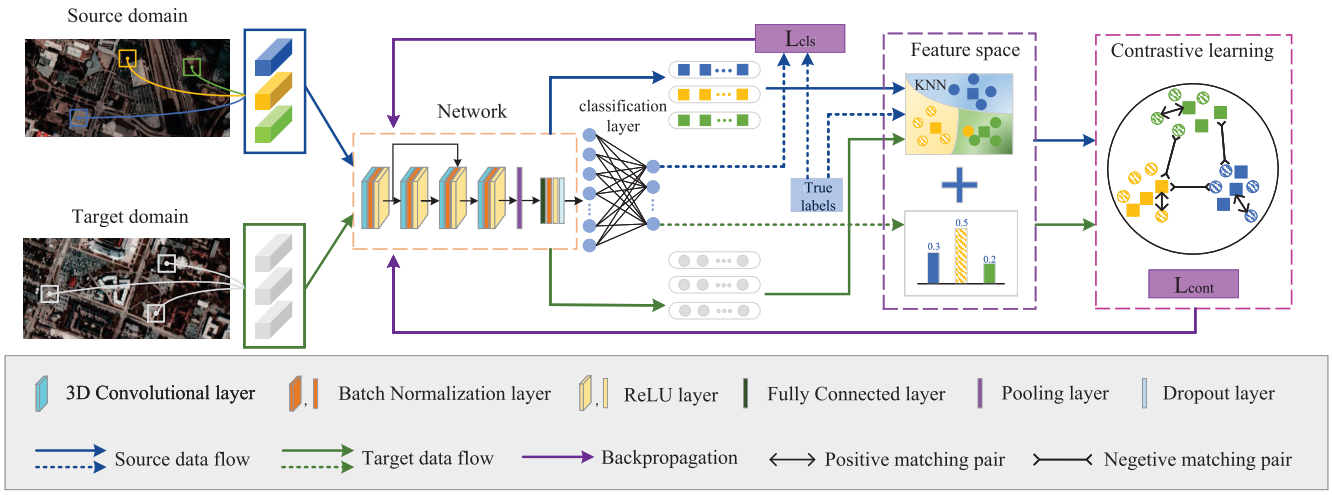


Fig. 1. Framework of the CLCM method. It mainly contains three modules, which are feature extraction network $\mathcal{G}$ (orange dotted line), high-confidence target samples selection (purple dotted line), and instance-level contrastive learning (magenta dotted line). Specifically, samples from two domains are put into the network $\mathcal{G}$ simultaneously, and the feature vectors corresponding to the samples are obtained. Then, the KNN classifier is used to filter out high-confidence target samples. Finally, domain shift is reduced by implementing CDCL.

and the K-nearest neighbor (KNN) classifier, the consistent samples are dropped into the network for contrastive learning. To a certain extent, the learning error caused by misclassified samples is reduced.

In summary, the main contributions of the proposed CLCM are summarized as follows.

1) A novel CDCL HSIC framework is proposed, which can effectively handle domain shift. Good performance is obtained on three cross-scene classification tasks.
2) A category information-based strategy is designed to construct positive and negative matching pairs across domains in the instance-level contrastive learning framework. Using contrastive loss, domain-invariant and class-discriminative features can be simultaneously extracted.
3) According to the predicted labels obtained by different strategies, high-confidence target samples can be selected for contrastive learning. Meanwhile, simple 3-D convolutions effectively extract spectral and spatial fusion features of HSIs. Hence, the quality of class-discriminative feature learning is further improved.

The rest of this article is organized as follows. Section II details our proposed CLCM method. Experimental results and analysis are shown in Section III. Finally, Section IV concludes this article.

II. Proposed Method

The unsupervised DA techniques aim to transfer a learned model from a labeled source domain to a related unlabeled target domain. In this article, given a labeled source domain with n_s sample and label pairs $\mathcal{D}_s = (\mathcal{X}_s, \mathcal{Y}_s) = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$, and an unlabeled target domain with n_t samples $\mathcal{D}_t = \mathcal{X}_t = \{x_j^t\}_{j=1}^{n_t}$, where $y_i^s \in \mathcal{L} = \{1, 2, \ldots, C\}$ and C is the number of classes, the objective is to train a model on $\mathcal{D}_s \cup \mathcal{D}_t$ to assign precise labels to $\mathcal{D}_t$. The model consists of a feature extraction network $\mathcal{G}$ and a task-specific classification layer $\mathcal{F}$ that classifies features into C categories.

Fig. 1 shows the framework of the proposed CLCM. Its essence is to extract features through the network $\mathcal{G}$ and perform instance-level contrastive learning to reduce domain shift.

A. Feature Extraction Network

In deep learning, the CNN is a common framework used to extract the spectral–spatial information of adjacent pixels in HSIs. Generally, the network architecture consists of convolution, pooling, and activation functions. In this work, we mainly use 3-D convolution as the basic convolution operation for building the feature extraction network $\mathcal{G}$. Let the HSI data be denoted as $\mathcal{H} \in \mathcal{R}^{h \times w \times d}$, where h (w) is the height (width) of the spatial dimension and d is the number of spectral bands. During training, some 3-D cubes of size $p \times p \times d$ centered on a pixel are fed into the network $\mathcal{G}$ as input, whose land-cover class is determined by the center pixel. Here, p represents the patch_size. Usually, the height and width of 3-D cubes are equal. Next, some $7 \times 7 \times 48$ cubes obtained from the Houston dataset are used as an example to illustrate the network structure.

Specifically, the network $\mathcal{G}$ consists of two 3-D convolutions, a residual block, a pooling layer, and a fully connected (FC) layer, as shown in Fig. 2. After each convolution operation, it is followed by a batch norm (BN) layer and a ReLU activation function. In the network $\mathcal{G}$, the first convolution is 24 $1 \times 1 \times 7$ kernels with stride (1, 1, 2), extracting spectral information while removing redundant spectral bands and generating 24 more condensed 3-D cube of size $7 \times 7 \times 21$. The following residual block focuses on learning more efficient spectral feature representations and maps the input into 24 $7 \times 7 \times 21$ cubes. In the residual block, all convolution operations go through the padding to ensure that the output cube size is consistent with the input size. The last

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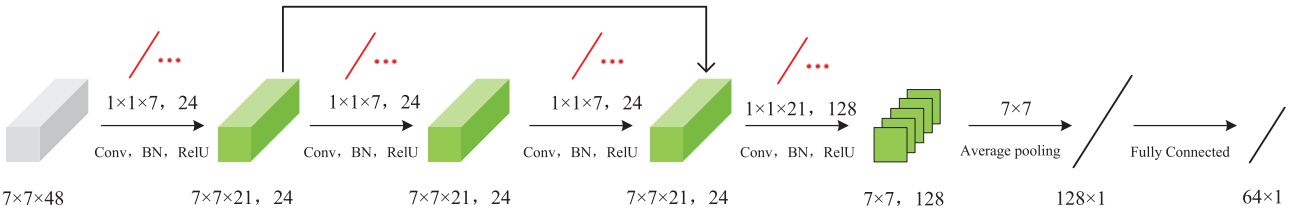


Fig. 2. Detailed structure of the feature extraction network $\mathcal{G}$.

convolutional layer contains 128 $1 \times 1 \times 21$ spectral kernels, resulting in 128 7×7 2-D feature patches. By focusing on the learning of pixel spectral features, while taking into account the class consistency around pixels, an average pooling layer of size 7×7 is applied to each channel and results in a 128-D feature vector. Also, notice that the kernel size of the average pooling layer is the same as the patch_size of the initial input cube, which varies with the patch_size. Finally, the FC layer and a classification layer are used to generate the prediction results for $\mathcal{D}_t$.

In fact, there are various ways to build feature extraction networks [31], [32], [33]. But for HSIs, 3-D convolutions can effectively extract spectral and spatial information. The proposed network architecture first focuses on spectral information through convolution with a window of 1×1 and then preserves spatial information through global average pooling. Finally, the spectral–spatial fusion features are obtained.

B. Contrastive Learning

1) Revisit of Contrastive Learning: Contrastive learning is to promote compactness within classes and discretization between classes and then learning common and discriminative features. Specifically, in the field of CV, the common practice is to record a given picture as an anchor image, and the pictures after data augmentation (e.g., random Gaussian blur, random color distortion, random picture cutting, etc.) are used as positive matching samples and all the rest pictures in a batch are negative matching samples. Even if there may be samples of the same type as the anchor in a batch. The InfoNCE loss is often used in advanced contrastive learning architectures [27], which aims to make positive matching pairs closer in representation space while negative matching pairs are farther apart. The definition of contrastive loss is

$$\mathcal{L} = -\log \frac{\exp\left(\psi\left(q, k^{+}\right)/\tau\right)}{\sum_{i=0}^{K} \exp(\psi(q, k_i)/\tau)} \tag{1}$$

where q is an anchor, k^{+} is a positive matching sample related to q, and the summation over the denominator consists of one positive and K negative matching samples. Here, τ is a temperature parameter. In addition, ψ represents the score function to measure the similarity between q and k_i, such as the commonly used cosine similarity or Euclidean distance [34]. To make the learned features better aligned in the embedding space, the normalized dot product is adopted, which has proven to effectively improve the representation quality [35]

$$\psi(q, k) = \frac{q^{\top} k}{\|q\| \|k\|}. \tag{2}$$

In short, the key to contrastive learning is to construct appropriate positive and negative matching pairs. Existing works on self-supervised contrastive learning force the augmented views of anchor images as positive matching samples, using a large batch size [26] or memory bank [36] as a source of negative matching samples to address this problem. All of these methods teach the model to recognize images of the same class from different augmented views and distinguish between images of different classes, thereby learning invariant and discriminative features. However, in the unsupervised DA problem, the distribution of $\mathcal{D}_s$ and $\mathcal{D}_t$ is different, and there has a large domain difference. The construction of positive and negative matching pairs requires careful consideration.

2) CDCL Based on Category Matching: Drawing inspiration from [28], it is desired to construct positive and negative matching pairs across domains, with category information as an entry point. In the embedding space, it is assumed that cross-domain samples from the same class are all clustered together, while cross-domain samples from different classes are separated, no matter if it is from $\mathcal{D}_s$ or $\mathcal{D}_t$. As shown in Fig. 3, there is a large domain gap between $\mathcal{D}_s$ and $\mathcal{D}_t$, even if the sample distribution of the same class is completely inconsistent in the two domains. Our original intention is to connect two domains through contrastive learning in the feature space to reduce domain differences and achieve feature matching. Different from [28], only the source domain is considered as an anchor in our work. Because in the unsupervised DA case, the source domain has real labels and the target domain is unlabeled. If the target samples with pseudo-labels are recognized as anchors, once the pseudo-label is wrong, the matched positive and negative samples are not correct. This will not be conducive to network learning. Specifically, the black arrow indicates that the positive matching samples of the anchor samples in $\mathcal{D}_s$ are found in $\mathcal{D}_t$, and they are expected to attract each other. The red arrows indicate its negative matching samples, which are expected to be mutually exclusive. Although the construction method of this contrastive learning is simple, it can connect two domains with different distributions and then learn robust and class discriminative features. Besides, the category information of the two domains needs to be provided. In the training process, the source domain has real labels. Regarding $\mathcal{D}_t$, we use the pseudo-label predicted by the network here, namely

$$\hat{y_j^t} = \operatorname{argmax}_{\mathcal{F},\mathcal{G}}\left(\mathcal{F}\left(\mathcal{G}\left(x_j^t\right)\right)\right). \tag{3}$$

Based on the above analysis, we construct a category matching-based contrastive learning framework for the DA problem of HSIC, using the features extracted by the 3-D CNN to directly align two domains. Specifically, a sample x_i^s for the cth category in $\mathcal{D}_s$ is recorded as an anchor, all samples with

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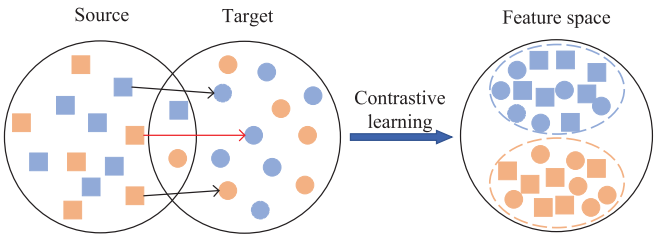


Fig. 3. Schematic of the proposed contrastive learning. The squares (circles) represent the source (target) samples, and the two categories are represented by orange and blue colors, respectively. Among them, the black (red) line represents the mutual attraction (repulsion) of the samples.

predicted label c in $\mathcal{D}_t$ are recorded as its positive matching samples, and all samples with predicted category from $C\backslash c$ in $\mathcal{D}_t$ are regarded as its negative matching samples. Unlike other contrastive losses, the denominator in this article only computes the sum of the similarities between the anchor and all samples from different classes (negative). The contrastive loss corresponding to the sample x_i^s is defined as

$$\mathcal{L}_{\text{cont}}^{x_i^s} = -\frac{1}{\left|I_{z_i^s}^{t+}\right|} \sum_{z_i^{t+} \in I_{z_i^s}^{t+}} \log \frac{\phi\left(z_i^s, z_i^{t+}\right)}{\phi\left(z_i^s, z_i^{t+}\right) + \sum_{z_i^{t-} \in I_{z_i^s}^{t-}} \phi\left(z_i^s, z_i^{t-}\right)} \tag{4}$$

$$\phi(a, b) = \exp\left(\frac{\psi(a, b)}{\tau}\right) \tag{5}$$

where z_i^s and z_i^t are the features extracted by $\mathcal{G}$ corresponding to the samples x_i^s and x_i^t, respectively, that is, $z_i^s = \mathcal{G}(x_i^s)$ and $z_i^t = \mathcal{G}(x_i^t)$. Besides, $I_{z_i^s}^{t+}$ ($I_{z_i^s}^{t-}$) represents the set of positive (negative) matching samples in $\mathcal{D}_t$ with the same (different) category as z_i^s, and $|I_{z_i^s}^{t+}|$ ($|I_{z_i^s}^{t-}|$) denotes the number of positive (negative) matching samples in the set, respectively. More importantly, the contrastive loss on sample x_i^s is the average of multiple pairs of positive matching samples because there are generally more than one positive matching sample pair in the target domain. Considering that the final contrastive loss needs to traverse every sample in the source domain $\mathcal{D}_s$, it is defined as

$$\mathcal{L}_{\text{cont}} = \frac{1}{n_s} \sum_{x_i^s \in \mathcal{X}_s} \mathcal{L}_{\text{cont}}^{x_i^s}. \tag{6}$$

C. High-Confidence Sample Selection (HCSS)

In unsupervised DA, the labels of $\mathcal{D}_t$ are not available. However, in CDCL, the essence is that no matter whether the samples come from $\mathcal{D}_s$ or $\mathcal{D}_t$, in the embedding space, samples of the same class can be tightly clustered, while samples of different classes are separated widely. Based on this, the goal of learning the common feature representation between the same class and distinguishing different classes is achieved. Therefore, category information is indispensable and critical when conducting contrastive learning. The ground truths of the source domain $\mathcal{Y}_s$ are available while considering the target domain, we can only utilize the pseudo-labels of the target samples. For example, some sophisticated methods annotate the target domain by constructing an affinity matrix or a probabilistic annotation matrix [37], [38]. Since the target pseudo-label is noisy, the accuracy of the predicted pseudo-label affects the learned feature representation.

In this regard, we consider filtering out high-confidence target samples for contrastive learning. Through the probability assigned to each sample by network training, a target pseudo-label can be obtained, as shown in (3). In addition, the 1-NN classifier is exploited to obtain another possible pseudo-label of the target sample, namely

$$(\mathcal{Z}_s, \mathcal{Z}_t, \mathcal{Y}_s) \to \tilde{\mathcal{Y}}_t$$

where $\mathcal{Z}_s = \{z_1^s, \ldots, z_N^s\}$ and $\mathcal{Z}_t = \{z_1^t, \ldots, z_N^t\}$ denote the features extracted by the network $\mathcal{G}$, $\tilde{\mathcal{Y}}_t = \{y_1^{\tilde{t}}, \ldots, y_N^{\tilde{t}}\}$ and $\hat{\mathcal{Y}}_t = \{y_1^{\hat{t}}, \ldots, y_N^{\hat{t}}\}$ are the target prediction pseudo-labels obtained by the two strategies. Here, N is the mini-batch size. Then, the two possible category information of $\mathcal{D}_t$ are compared, resulting in

$$\delta_j = \begin{cases} 1, & y_j^{\tilde{t}} = y_j^{\hat{t}} \\ 0, & y_j^{\tilde{t}} \neq y_j^{\hat{t}}. \end{cases} \tag{7}$$

Only the target samples with the same prediction results are screened out, that is, samples in $\bar{\mathcal{Z}}_t = \{z_j^t | \delta_j z_j^t \neq 0, j = 1, \ldots, N\}$ participate in contrastive learning. Then the positive and negative matching samples corresponding to the source samples (anchor) are selected from the filtered target samples set $\bar{\mathcal{Z}}_t$.

D. Total Loss Function

Here, the basic idea is to implement instance-level contrastive learning to tackle the DA issue. The overall loss of the entire framework only includes the classification loss of $\mathcal{D}_s$ and the cross-domain contrastive loss, described as

$$\mathcal{L} = \mathcal{L}_{\text{cls}}^s + \lambda \mathcal{L}_{\text{cont}} \tag{8}$$

$$\mathcal{L}_{\text{cls}}^s = \frac{1}{n_s} \sum_{i=1}^{n_s} J\left(\mathcal{F}\left(\mathcal{G}\left(x_i^s\right)\right), y_i^s\right) \tag{9}$$

where J is the cross-entropy loss function and λ is the tradeoff parameter. The pseudo-code of CLCM is summarized in Algorithm 1.

III. Experimental Results

A. Data Description

In this section, three HSI datasets are used to verify the effectiveness of the proposed CLCM algorithm, namely the Houston dataset, the HyRank dataset, and the Hangzhou-Shanghai dataset.

1) Houston: The task contains the Houston 2013 scene [39] and the Houston 2018 scene [40], which are images of the University of Houston campus and surrounding area in different years. The Houston 2013 scene contains 349×1905 pixels and 144 spectral bands. The Houston 2018 scene only contains 48 spectral bands. To maintain consistency, we extract the 48 spectral bands corresponding to the Houston 2018 scene from the Houston 2013 scene. Experiments are conducted using seven shared categories, and details are shown in Table I.

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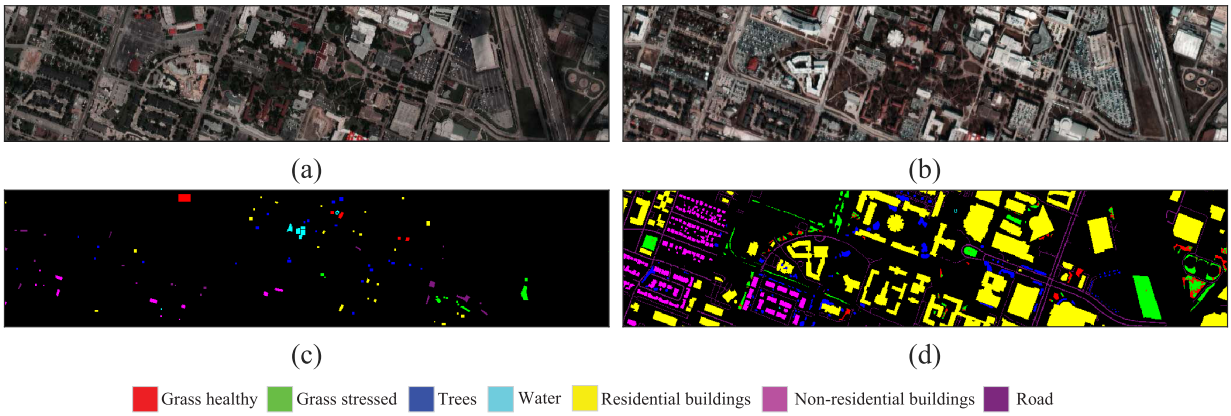


Fig. 4. Houston image. Pseudo-color image of (a) Houston 2013 and (b) Houston 2018. Ground-truth map of (c) Houston 2013 and (d) Houston 2018.

Algorithm 1 CLCM

Input: The source data $\mathcal{D}_s = \{x_i^s, y_i^s\}_{i=1}^{n_s}$, the target data $\mathcal{D}_t = \{x_j^t\}_{j=1}^{n_t}$, the iterations per epoch R, the total epoch number E, the trade-off parameter λ.
Output: Target predicted labels Y_t.

Begin

Randomly initialize network $\mathcal{G}$.
for $e = 1$ to E **do**
  for $r = 1$ to R **do**
    Select high-confidence target samples.
    Perform contrastive learning by Eq. (4).
    Back-propagate and update network $\mathcal{G}$ by Eq. (8).
  end for
end for
Obtain the target pseudo labels Y_t.

TABLE I
LAND COVER CLASS AND CORRESPONDING SAMPLES NUMBERS ON THE HOUSTON DATASET

Class		Number	
No.	Name	Houston 2013 (source)	Houston 2018 (target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

TABLE II
LAND COVER CLASS AND CORRESPONDING SAMPLE NUMBERS ON THE HYRANK DATASET

Class		Number	
No.	Name	Dioni (source)	Loukia (target)
1	Dense Urban Fabric	1262	206
2	Mineral Extraction Sites	204	54
3	Non Irrigated Arable Land	614	426
4	Fruit Trees	150	79
5	Olive Groves	1768	1107
6	Coniferous Vegetation	361	422
7	Dense Sderophyllous Vegetation	5035	2996
8	Sparce Sderophyllous Vegetation	6374	2361
9	Sparcely Vegetated Areas	1754	399
10	Rocks and Sand	492	453
11	Water	1612	1393
12	Coastal Water	398	421
Total		20024	10317

TABLE III
LAND COVER CLASS AND CORRESPONDING SAMPLE NUMBERS ON THE HANGZHOU-SHANGHAI DATASET

Class		Number	
No.	Name	Hangzhou (source)	Shanghai (target)
1	Water	18043	123123
2	Land/building	77450	161689
3	Plant	40207	83188
Total		135700	368000

2) HyRank: The HyRank dataset has been developed in the framework of the ISPRS Scientific Initiatives. It was collected by the Hyperion sensors. In fact, it contains a total of five HSIs of 176 bands, of which the Dioni and Loukia datasets are used for the experiments. They have 250×1376 pixels and 249×945 pixels, respectively, with 12 consistent categories, as shown in Table II.

3) Hangzhou-Shanghai: The two datasets for this task were acquired by the EO-1 Hyperion hyperspectral sensor. After removing some bad bands, there are 198 spectral bands left [10]. The Hangzhou (Shanghai) image contains 590×230 (1600×230) pixels with three categories, namely roads/buildings, plants, West Lake, and Qiantang River basin (the waters of the Yangtze River and Huangpu River). The details of land cover classes are shown in Table III.

The pseudo-color images and ground-truth maps are shown in Figs. 4–6.

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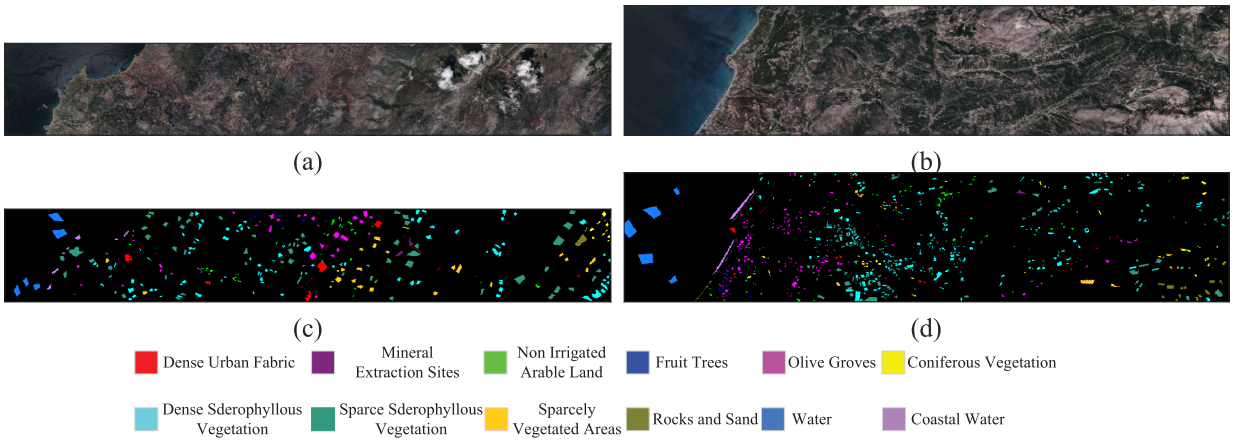


Fig. 5. HyRank image. Pseudo-color image of (a) Dioni and (b) Loukia. Ground-truth map of (c) Dioni and (d) Loukia.

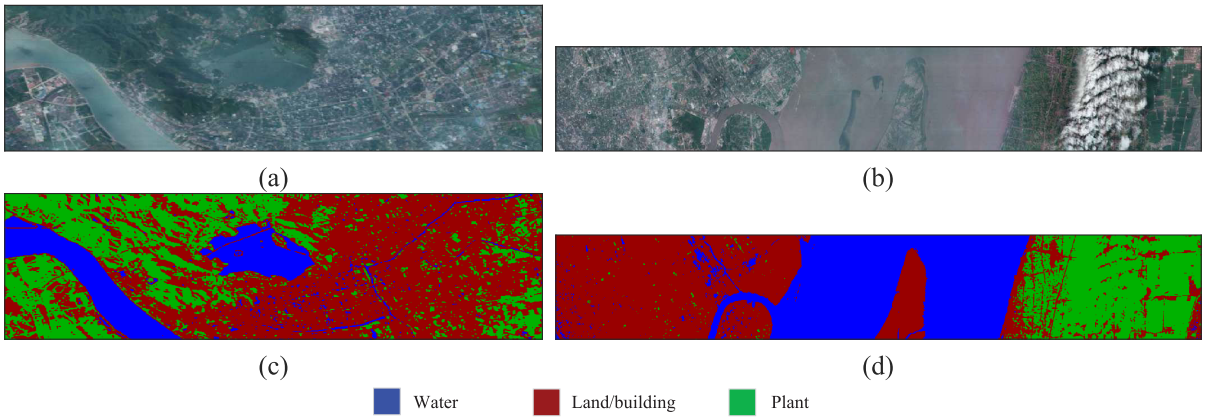


Fig. 6. Hangzhou-Shanghai image. Pseudo-color image of (a) Hangzhou and (b) Shanghai. Ground-truth map of (c) Hangzhou and (d) Shanghai.

B. Comparison Methods

The proposed algorithm is compared with other DA algorithms, including KNN (No Adaptation), SA [16], JDA [20], DANs [22], DSAN [23], MRAN [24], CLDA [25], and CDCL [28].

In the traditional DA algorithm here, the subspace dimension of SA and JDA is $k = 20$, and a 1-NN classifier is used. The patch_size of the deep DA algorithm is selected from $\{3, 5, 7, 9\}$, and settings are 7×7 for Houston tasks, 5×5 for the HyRank task, and 3×3 for the Hangzhou-Shanghai task in the comparison algorithms. The network is trained using the mini-batch SGD method. And the choice of initial learning rate is in $\{1e^{-1}, 1e^{-2}, 1e^{-3}, 1e^{-4}, 5e^{-5}\}$. The number of training iterations is 120 on three tasks. Moreover, the batch size is uniformly set to 64. The temperature parameter τ is set to 0.5 for the CDCL and the proposed CLCM methods. In fact, some scholars have also shown that the role of the temperature coefficient is to adjust the degree of attention to difficult samples [41]. Smaller temperature coefficients focus more on separating this sample from the most similar other samples. Therefore, when the temperature coefficient takes an appropriate value, it has uniformity and tolerance for all samples, and the performance of the model produced by this loss remains basically stable. Usually, it generally takes a value of 0.5.

Considering the categories of the HyRank dataset and its imbalance, the spectral curves of some categories are very similar. Without adding more computational overhead, we perform data augmentation rotation on some categories (Classes 2, 4, 6, and 10) of its source domain. Specifically, we rotate 3-D cubes by 90°, 180°, and 270° as data enhancement samples. These are fed into the network training along with the original cubes. The amount of training data for these categories is equivalent to four times the amount of original data. The overall accuracy (OA) and the kappa coefficient (κ) are used to evaluate the classification performance.

C. Experiments

1) Experiments on the Houston Task: The Houston dataset is the imaging of the University of Houston and the surrounding environment in different years, and the sensors also change over time. So, the two scenes of Houston 2013 and Houston 2018 are very different, which makes the classification very difficult.

Table IV provides the classification performance of different DA algorithms on the Houston 2018 dataset. It demonstrates that the classification accuracy of the simplest deep DA algorithms (e.g., DAN, DSAN, and MRAN) is even worse than the KNN method. This is because most of their basic frameworks are the general ResNet-50, which cannot fully consider

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TABLE IV
CLASSIFICATION ACCURACIES ON THE HOUSTON DATASET

Class	KNN	SA	JDA	DAN	DSAN	MRAN	CLDA	CDCL	CLCM
1	78.94	70.29	81.82	81.38	83.49	76.47	**88.03**	73.95	85.83
2	94.82	**96.17**	80.36	66.21	67.49	68.72	83.06	69.51	72.42
3	66.52	65.18	65.80	72.59	75.10	72.22	59.69	70.92	**75.75**
4	**100**	**100**	**100**	94.55	98.64	94.55	**100**	93.43	99.55
5	64.00	63.36	60.50	67.05	64.64	68.83	**83.92**	58.87	75.80
6	68.10	68.87	59.86	64.41	64.54	63.77	71.98	74.14	**87.03**
7	60.91	54.00	**72.07**	64.44	61.30	66.90	81.49	59.80	49.45
OA(%)	69.49	68.90	64.15	66.09	65.48	65.89	75.12	70.30	**79.45**
κ	0.539	0.525	0.487	0.509	0.505	0.509	0.628	0.545	**0.662**

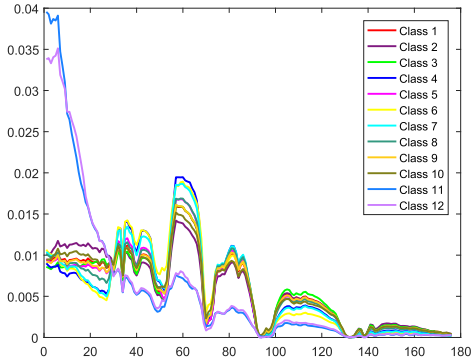


Fig. 7. Mean spectrum of all classes on the HyRank dataset.

TABLE V
CLASSIFICATION ACCURACIES ON THE HYRANK DATASET

Class	KNN	SA	JDA	DAN	DSAN	MRAN	CLDA	CDCL	CLCM
1	15.05	14.56	20.14	9.85	10.29	6.17	12.62	**37.48**	16.84
2	51.85	50.00	**92.54**	75.00	35.74	45.19	22.22	59.81	72.41
3	36.15	35.21	**75.65**	32.21	30.09	27.49	37.09	39.60	27.37
4	0	0	69.62	36.08	55.19	7.59	0	**76.84**	4.30
5	0.18	0.18	29.26	13.46	25.13	12.66	5.06	20.25	**58.24**
6	27.96	24.88	5.80	8.20	**29.69**	14.12	1.66	4.19	17.49
7	80.37	80.57	67.23	69.00	69.16	69.05	**85.65**	76.93	81.49
8	33.59	41.72	63.22	74.96	71.38	73.58	**80.94**	70.34	78.01
9	31.08	27.57	18.32	38.80	49.10	52.08	3.76	46.99	**69.20**
10	**40.84**	40.40	40.25	1.24	6.45	3.73	8.17	39.29	0.20
11	**100**	**100**	**100**	97.85	98.28	97.98	**100**	98.04	**100**
12	**100**	**100**	**100**	95.99	99.36	90.95	98.34	99.79	99.81
OA(%)	54.83	56.41	61.11	59.85	61.89	59.50	63.92	64.92	**70.62**
κ	0.453	0.468	0.529	0.505	0.537	0.499	0.542	0.574	**0.638**

TABLE VI
CLASSIFICATION ACCURACIES ON THE HANGZHOU-SHANGHAI DATASET

Class	KNN	SA	JDA	DAN	DSAN	MRAN	CLDA	CDCL	CLCM
1	93.35	**94.24**	91.66	91.85	92.43	92.63	87.88	93.43	88.73
2	41.51	52.06	72.88	67.52	68.56	68.42	93.21	82.33	**96.06**
3	**100**	**100**	91.27	**100**	**100**	99.35	98.50	99.45	96.40
OA(%)	72.07	77.01	83.32	83.00	83.65	83.51	92.62	89.92	**93.69**
κ	0.599	0.665	0.748	0.747	0.756	0.754	0.886	0.847	**0.902**

the characteristics of HSIs and cannot effectively extract spectral–spatial features. The classification performance of the CDCL algorithm is better than the traditional DA algorithms and the basic deep DA algorithms, which can reach 70.30%, even though its network architecture is still ResNet-50. This means that compared with directly minimizing the MMD distance in the feature space, using contrastive learning for common features between the samples from the same class can better achieve cross-domain classification performance. The proposed CLCM uses 3-D convolution as the basic element of the network architecture to effectively extract the spectral information of HSI and perform contrastive learning in the embedding feature space, which can obtain a relatively good classification result, that is, 79.45%.

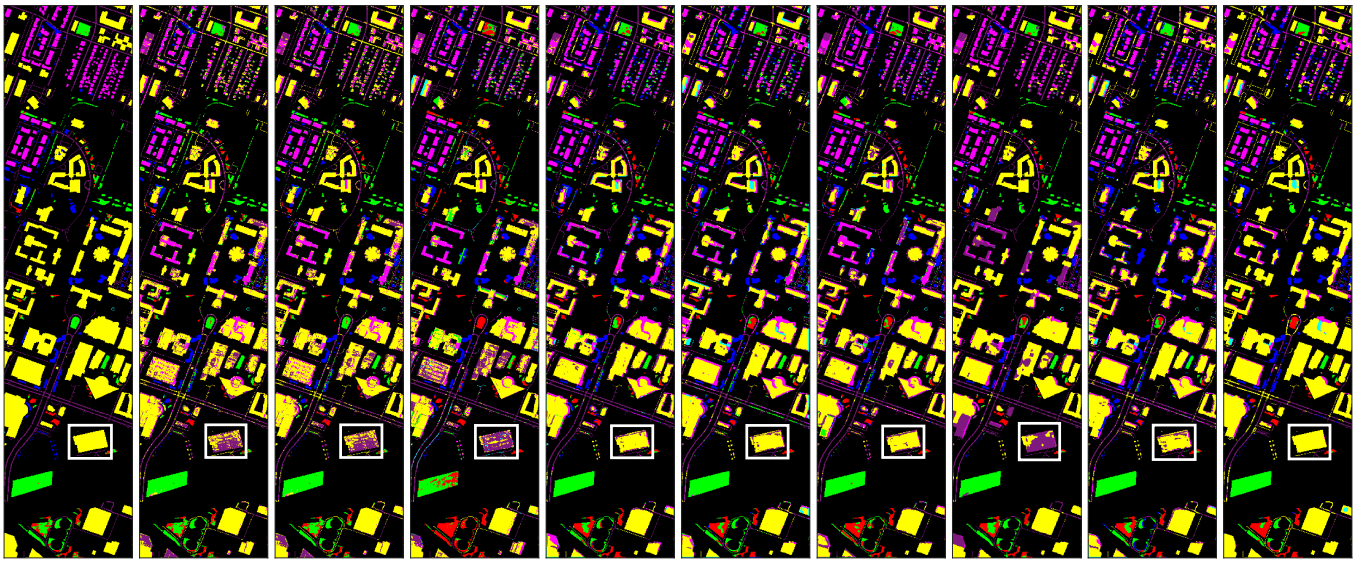
2) Experiments on HyRank Task: There are 12 common categories for Loukia and Donio datasets. Due to the similarity between categories, cross-scene classification is very difficult, so all algorithms perform poorly on the HyRank task. For example, the fourth ("Fruit Trees") and fifth ("Olive Groves") categories are trees, and the sixth ("Coniferous Vegetation"), seventh ("Dense Sderophyllous Vegetation"), and eighth ("Sparce Sderophyllous Vegetation") categories are plants, and so on.

Table V shows the specific classification results of different DA methods on the HyRank task. It can be seen that all DA algorithms are very accurate in the classification of the 11th ("Water") and 12th ("Coastal Water") categories, basically more than 95%, or nearly 100%. Moreover, the CDCL algorithm is still superior to the basic deep DA algorithms on the whole, which is 3%–5% higher. The CLCM maximizes the similarity between positive matching pairs and minimizes the similarity between negative matching pairs through contrastive learning, to learn robust and discriminant features. Its accuracy can reach 70.62%, nearly 6% higher than the CDCL algorithm. In particular, CLCM can still effectively ensure the classification accuracy of the 11th ("Water") and 12th ("Coastal Water") categories, which are 100% and 99.81%, respectively. As instance-level contrastive learning is used in this article, when there are few samples given and the categories are easily confused, the classification results of some categories are not ideal. For example, the classification accuracy of the fourth (Fruit Trees) and tenth ("Rocks and Sand") categories are only 4.3% and 0.2%, respectively. In fact, Fig. 7 shows the mean spectral curves for all classes on the HyRank dataset. It can be seen intuitively that many categories are very similar, that is, different classes of substances have similar spectral features. And in the case of DA, when there are differences between the two domains, the classification of these categories becomes more difficult.

3) Experiments on the Hangzhou-Shanghai Task: There are only three common categories for this task, namely water, roads/buildings, and plants. Three categories are relatively easy to separate, so all DA algorithms have better classification results.

Table VI provides the classification performance of different DA algorithms on this task. On the whole, only using KNN, the classification result can reach 72.07%. SA can be improved

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Fig. 8. Houston task. (a) Ground truth. (b) KNN (69.49%). (c) SA (68.90%). (d) JDA (64.15%). (e) DAN (66.09%). (f) DSAN (65.48%). (g) MRAN (65.89%). (h) CLDA (75.12%). (i) CDCL (70.30%). (j) CLCM (79.45%).

by 5% by learning a linear transformation for SA. The algorithms JDA, DAN, DSAN, and MRAN use the MMD metrics to reduce domain distribution differences, which can basically reach about 83%. However, the CDCL and CLCM algorithms using contrastive learning are still superior to optimizing statistical distribution differences, which can obtain 89.92% and 93.69%, respectively. Moreover, they substantially improve the separation of the second ("Land/building") category, while basically guaranteeing the accurate classification of the first ("Water") and third ("Plant") categories. The classification performance of CDCL in the second ("Land/building") category is also improved by nearly 10% compared to the highest result (i.e., JDA), and CLCM is improved by nearly 14% on the basis of CDCL.

Overall, the deep DA methods are still superior to the traditional DA methods, and the classification performance is generally better. More importantly, it can automatically extract features and continuously update the network. To realize domain migration, DAN, DSAN, and MRAN add an adaptive layer in the network and use the MMD matrix or its variants to reduce the domain difference, which can effectively improve performance. While CDCL is still the premise of ResNet-50 in the entire framework, the addition of contrastive learning effectively improves performance on three tasks. This indicates that contrastive learning may be better at extracting domain-invariant and discriminative features than directly reducing statistical distribution differences. However, our proposed CLCM applies contrastive learning and considers using the network suitable for HSIs to extract spectral and spatial fusion features. On this basis, the classification effect is further improved. In addition to the above comparative experimental data results, we also visually display the classification prediction maps on the target domain in Figs. 8–10. Obviously, the classification map of our CLCM is more similar to the ground-truth map. As highlighted, the classification map inside the rectangular box is significantly better than other comparison methods.

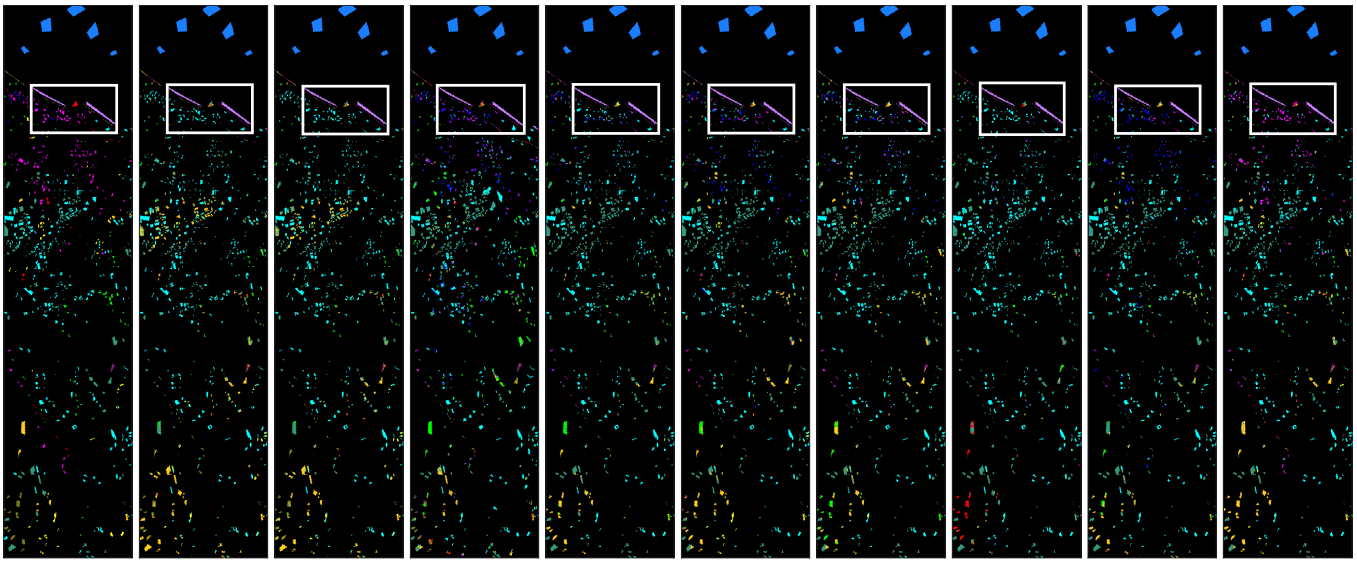
D. Ablation Analysis

In this section, to verify whether each module in CLCM is effective, ablation experiments are performed on the Houston dataset. In this regard, we design four experiments, and the results are shown in Table VII.

When no DA method is added and only the classification loss of $\mathcal{D}_s$ is used, the accuracy is the worst, only 63.02%. This means that when there is a domain shift, directly transferring the model trained on the labeled source data to the unlabeled target domain is infeasible. In experiment 2, instance-to-instance contrastive learning is added. Regardless of whether the samples are source samples or target samples, as long as they come from the same class, their similarity will be strengthened to make them cluster together, and samples of different classes need to weaken their similarity and make them distant. Meanwhile, the training of source data is ensured through cross-entropy loss. The classification performance can be significantly improved by nearly 15%–77.92%. It is shown that contrastive learning can indeed effectively achieve cross-scene classification, even with domain shifts. When experiment 3 replaces the prediction output by the network softmax layer with the prediction obtained by the 1-NN classifier and implements contrastive learning, the classification result increases to 78.52%.

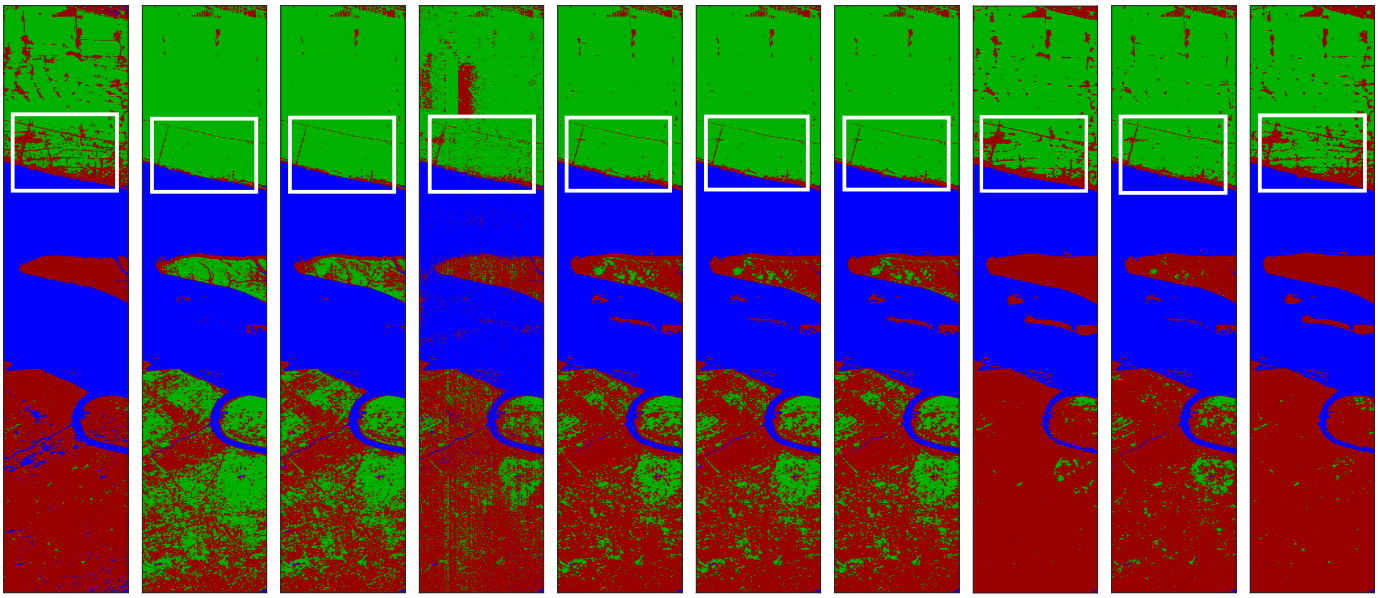
Finally, the proposed CLCM uses contrastive learning to connect two domains to learn invariant and discriminative features and uses HCSS to ensure that the selected target samples have high confidence. In this way, the interference of misclassified samples on instance-level contrastive learning can be reduced, and finally, a relatively good classification performance can be obtained, that is, 79.45%.

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Fig. 9. HyRank task. (a) Ground truth. (b) KNN (54.83%). (c) SA (56.41%). (d) JDA (61.11%). (e) DAN (59.85%). (f) DSAN (61.89%). (g) MRAN (59.50%). (h) CLDA (63.92%). (i) CDCL (64.92%). (j) CLCM (70.62%).



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 10. Hangzhou-Shanghai task. (a) Ground truth. (b) KNN (72.07%). (c) SA (77.01%). (d) JDA (83.32%). (e) DAN (83.00%). (f) DSAN (83.65%). (g) MRAN (83.51%). (h) CLDA (92.62%). (i) CDCL (89.92%). (j) CLCM (93.69%).

TABLE VII
ABLATION STUDY OF CLCM FOR THE HOUSTON TASK

	$\mathcal{L}_{cls}^{s}$	$\mathcal{L}_{cont}$	KNN	HCSS	OA(%)
1	✓				63.02
2	✓	✓			77.92
3	✓	✓	✓		78.52
CLCM	✓	✓	✓	✓	**79.45**

E. Parameter Analysis

In CLCM, the tradeoff parameters λ and patch_size are worth discussing. For the tradeoff parameter λ, Fig. 11 illustrates the variation trend of OA on three cross-domain tasks when λ varies on $[0.01, 1]$. Obviously, when λ is equal to 0.1, the OA reaches the highest on the Houston task, and the fluctuation of the adjacent value around 0.1 is relatively stable on the three tasks. Thus, λ is always set to 0.1 during training to effectively reduce domain shift and achieve cross-scene HSIC.

Considering that different patch sizes may have an impact on the results, we choose different data input sizes from 3×3, 5×5, 7×7, and 9×9 for experiments. The results are shown in Fig. 12. The Houston task reaches its peak at size 7×7, and the HyRank and Hangzhou-Shanghai tasks achieve the best classification results at size 3×3. Therefore, choosing an appropriate patch_size in the experiment has a positive impact on the final performance.

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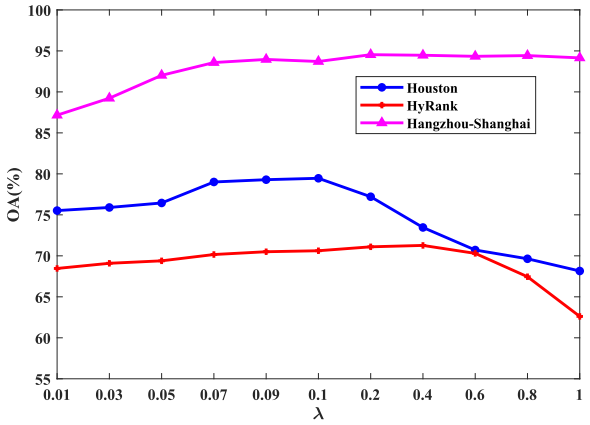


Fig. 11. OA of CLCM versus parameters λ on three tasks.

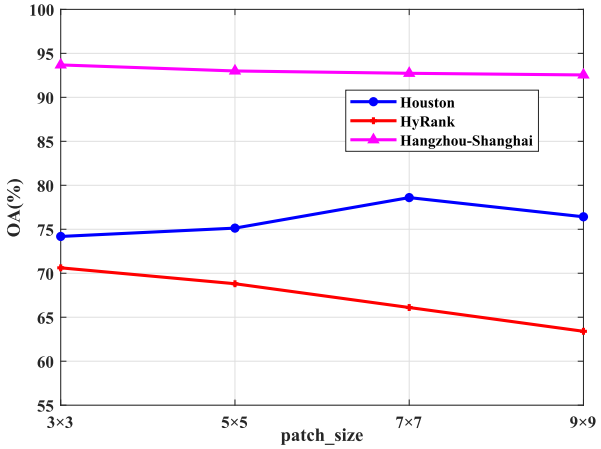


Fig. 12. OA of CLCM versus parameters patch_size on three tasks.

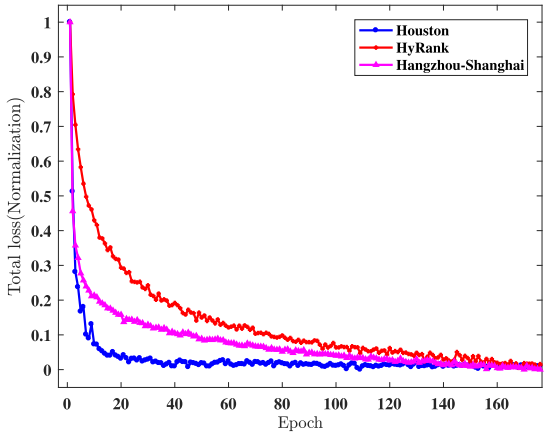


Fig. 13. Convergence analysis of CLCM on three tasks.

In addition, regarding the selection of the number of iterations, a convergence analysis is performed on the three tasks. It can be seen from Fig. 13 that the total loss eventually stabilizes as the network continues to train. Considering that continuing to increase the number of iterations will not affect the experimental results, it is possible to choose to terminate at an appropriate number of epochs to reduce computational overhead. In the experiment, the number of epochs is set as 120 for all three DA tasks.

TABLE VIII
ONE EPOCH TRAINING TIME (IN SECONDS) FOR DIFFERENT METHODS

methods	DAN	DSAN	MRAN	CLDA	CDCL	CLCM
time(s)	4.902	4.211	5.556	3.343	4.946	**2.711**

F. Feature Visualization

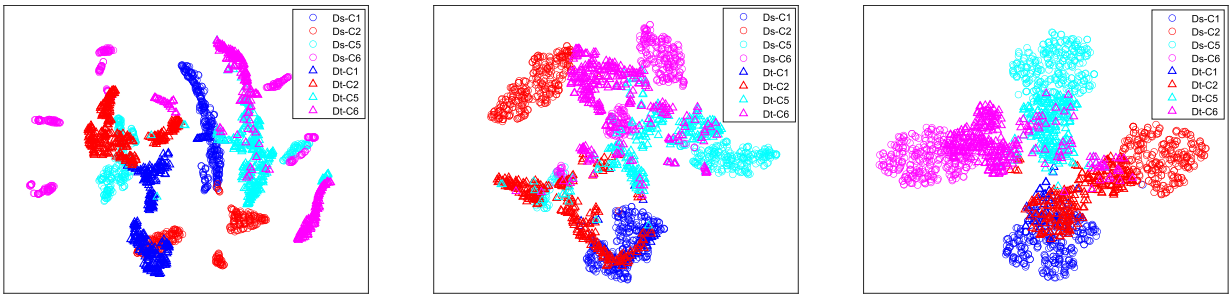
The strategy of DA is to reduce domain shift and realize domain migration. To visually observe the original data distribution of two domains as well as the aligned data distribution after using the DA method, the tSNE-based feature visualization results of the original data, CDCL, and the proposed CLCM method on the three classification tasks are shown in Figs. 14–16. It simultaneously displays the sample distribution of the classes chosen by two domains. Among them, the circles (triangles) represent the source (target) samples, respectively, and different colors represent different categories. On the Houston task, four confusing categories are selected, that is, the first ("Grass healthy") category, the second ("Grass stressed") category, the fifth ("Residential buildings") category, and the sixth ("Non-residential buildings") category. Four categories are also selected for the HyRank task: the seventh ("Dense Sderophyllous Vegetation") category, the eighth ("Sparce Sderophyllous Vegetation") category, the ninth ("Sparcely Vegetated Areas") category, and the 11th ("Water") category. The first three categories are prone to be confused. Finally, the Hangzhou-Shanghai task offers all three categories.

The Houston task is selected for feature visualization. From Fig. 14(a), it can be seen that the original data distribution is very chaotic. The two domains are very separated, and some categories are divided into several clusters, such as the magenta samples from the sixth category. After the CDCL method, whether it is a source sample or a target sample, those from the same class are clustered together to a certain extent. But the distance between some classes is still not well eliminated, such as the cyan samples from the fifth category in Fig. 14(b). However, after applying our CLCM algorithm, all samples from both domains are basically clustered into four clusters. Since some categories are difficult to distinguish, it is understandable that some samples are wrongly classified into other categories, such as red and blue samples in Fig. 14(c). Furthermore, on the other two datasets, both source and target samples from the same class are tightly clustered and separated from other classes. This fully demonstrates that contrastive learning in CLCM can effectively connect two domains and eliminate domain differences.

G. Analysis of the Computational Complexity

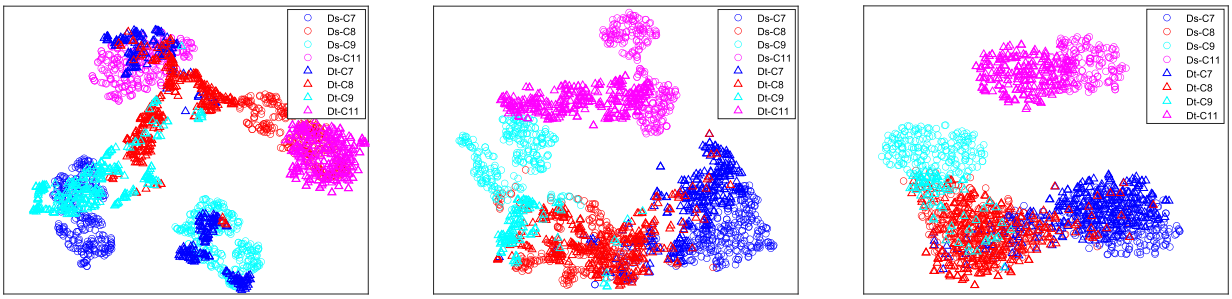
In this section, the computational complexity of different deep algorithms on the Houston task is evaluated. All experiments are performed under the PyTorch framework with a 3-GHz Intel Xeon Gold 6248R CPU and four NVIDIA GeForce RTX 2080 Ti GPUs. Table VIII shows the training time of different algorithms for one epoch, under the premise of the same parameter setting. Training of the proposed CLCM is fast due to the use of a very simple network. Considering

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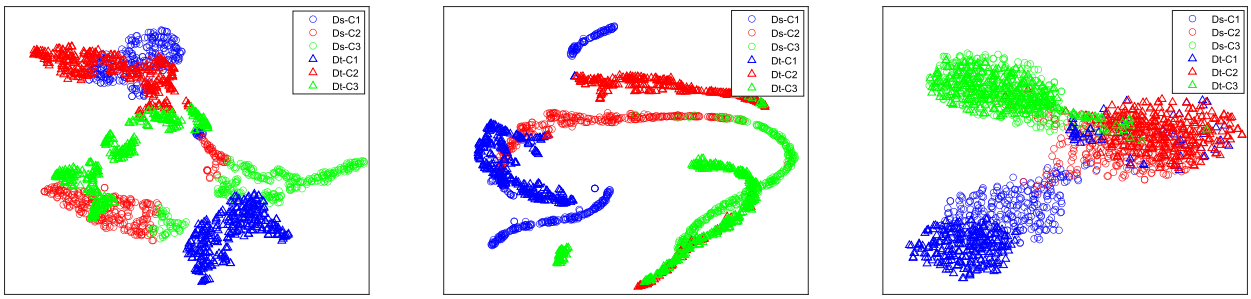
(a) (b) (c)

Fig. 14. tSNE feature visualization on the Houston task. (a) Original feature. (b) Features after CDCL alignment. (c) Features after CLCM alignment. The first, second, fifth, and sixth categories are denoted as blue, red, cyan, and magenta colors, respectively.



(a) (b) (c)

Fig. 15. tSNE feature visualization on the HyRank task. (a) Original feature. (b) Features after CDCL alignment. (c) Features after CLCM alignment. The seventh, eighth, ninth, and 11th categories are denoted as blue, red, cyan, and magenta colors, respectively.



(a) (b) (c)

Fig. 16. tSNE feature visualization on the Hangzhou-Shanghai task. (a) Original feature. (b) Features after CDCL alignment. (c) Features after CLCM alignment. The first, second, and third categories are denoted as blue, red, and green colors, respectively.

both training speed and classification performance, the proposed CLCM is a more efficient algorithm for cross-scene HSIC.

IV. Conclusion

In this article, a simple but novel contrastive learning based on a category-matching model is proposed for DA and classification of HSIs. In the proposed method, the source sample is used as an anchor, and its positive and negative matching samples are searched in $\mathcal{D}_t$. The two domains are associated with reduced domain discrepancy by forcing positive matching pairs to be closer and negative matching pairs to be farther away to perform instance-to-instance contrastive learning. To improve the quality of contrastive learning, on the one hand, we effectively extract the spectral–spatial features of HSIs through 3-D convolution. On the other hand, we consider HCSS and use classification-consistent samples in network training. Extensive experiments confirm the effectiveness of the proposed CLCM algorithm on three tasks.

In this work, high-confidence samples are screened out, focusing on sample-aware contrastive learning. However, there are still misclassified or confusing samples. To reduce the impact of noise samples on contrastive adaptation, we next consider further research on prototypical contrastive learning, where class centroids replace individual samples.

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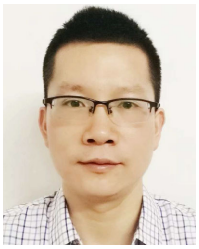
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